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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING(24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**AI Augmented DevSecOps: An Adaptive CI/CD Pipeline with Automated Vulnerability Triage**	
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Abstract

Modern CI/CD pipelines rely on multiple security tools — SonarQube for static analysis, OWASP ZAP for dynamic testing, and Trivy for vulnerability scanning — but these generate large volumes of findings with little prioritization, forcing developers to manually sift through alerts and often causing critical issues to be missed or release cycles to slow down. This project proposes an AI-Augmented DevSecOps framework that adds an intelligent triage layer to an adaptive CI/CD pipeline: alongside standard build, scan, and deployment stages, a lightweight machine learning model classifies and ranks vulnerabilities by severity, exploitability, and historical remediation data, giving developers a prioritized triage list instead of raw scanner output. The pipeline uses Git-based version control, Docker containerization, and integrates SonarQube, OWASP ZAP, and Trivy, with the AI model also informing build/deploy gating decisions based on computed risk. The goal is to show that AI-driven vulnerability triage can reduce manual review effort, speed up remediation, and strengthen security in adaptive software delivery pipelines.

Signature of the Guide HOD